Alleviating Sparsity to Enhance Group Recommendation with Cross-Linked Domain Model

Yui Chee Xuan[1], Rosmamalmi Mat Nawi[1](✉), Nurul Aida Osman[2], and Nur Ziadah Harun[1]

[1] Faculty of Computer Science and Information Technology, Tun Hussein Onn University of Malaysia, 86400 Parit Raja, Johor, Malaysia
gi220016@student.uthm.edu.my, {rosmamalmi,nurziadah}@uthm.edu.my
[2] Computer and Information Sciences Department and Centre for Research in Data Science, University Technology Petronas, 32610 Seri Iskandar, Perak Darul Ridzuan, Malaysia
nurulaida.osman@utp.edu.my

Abstract. As a new search paradigm emerges, users' perspectives on information searching shift from finding information to receiving it. Recommender systems (RS) are an emerging method of obtaining information. RS has succeeded in many conventional domains, such as social media and online video platforms like 'YouTube.' An insufficient number of user-item ratings triggers obstacles to individual RS and group recommender systems (GRS). Data sparsity becomes an issue as a result of this incompleteness. The accuracy of recommendations given to a group suffers when there is a lack of data within the group. It happens as a result of ineffective group formation, which usually involves individuals who possess sparse user profiles. Most of the research being done now concentrates on this problem after group formation. Assuming that handling data sparsity at the individual level would be more efficient, this study concentrated on data sparsity before the group formation process. The primary goal is to create a cross-domain technique with Linked Open Data (LOD) technology to ensure that problems with data sparsity can be addressed before the group formation process. The experiment of the proposed method leveraging LOD and cross-domain knowledge simultaneously in this study suggests that it achieved the lowest prediction errors compared to other experiments carried out in this study. A more accurate rating prediction can further alleviate data sparsity in the user-item matrix before group formation using user profiles. Hence, this research will enhance the quality of recommendations by reducing data sparsity in user profiles.

Keywords: Group recommender system · linked open data · data sparsity · cross-domain

1 Introduction

As social and group activities like vacationing, watching movies, traveling to scenic spots, and participating in sporting events produce a lot of information requirements, recommender systems (RS) for user groups are becoming increasingly popular.

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In addition, according to Felfernig et al. [1], group recommender system (GRS) is a relatively new technology with few known commercially practical applications compared to conventional RS. Despite the fact that there is still little study focused on recommendations to a group of users, Xu et al. [2] report that there has been a significant rise in the demand for such recommendation applications.

GRS still has difficulties making reliable recommendations when data is limited [3]. This situation emerged due to the association between the GRS groups and insufficient data. Group formation will not be effective if user preferences are missing from the user profile. Hence, highlighting the lack of information in the group profile is crucial to offering groups high-quality and pertinent recommendations [3, 4].

2 Literature Review

Most recent studies use cross-domain [5–7] and Linked Open Data (LOD) technologies [8] for individual RS, including reducing data sparsity [8–10]. However, our literature survey indicates that no research has been done focusing on the prior group formation in GRS that simultaneously employs the cross-domain approach and LOD technology. Moreover, the GRS studies that independently concentrate on cross-domain or LOD technology did not address the group formation of GRS.

To guarantee that data sparsity concerns can be minimized before the group formation procedure, adopting the technique via cross-domain integration using LOD technology is suggested. The purpose of this study is to ascertain whether using LOD technology in conjunction with a cross-domain strategy will improve the quality of recommendations made to the group.

The data sparsity issue is addressed between groups, as evidenced by the current GRS research studies that address it after group formation. However, this study hypothesizes that it would be more efficient if the data sparsity issue were minimized before the group formation process. While many methods have been proposed to overcome data sparsity, such as recursive filtering approaches [11], artificial neural network approaches [12], and data imputation approaches [13], GRSs have yet to receive much attention.

Table 1. Related works pertaining to cross-domain in group recommender system.

Author	Method	Dataset and Domain
Liang et al. [14]	"Hierarchical attention neural network-based cross-domain group recommendation method (HANCDGR)"	1. Mafengwo (tourism website) 2. Yelp (restaurant dataset) 3. CAMRa2011 (movie rating records of group members) 4. MovieLens1M 5. MovieLens25M 6. MovieLens-Simi
Richa and Bedi [6]		Tourism and its sub-domain comprise hotels, restaurants, and others

Table 1 highlights relevant studies that use cross-domain approaches with data from other relevant domains to alleviate data sparsity for group recommendations. We also provide a brief overview concerning the study's scope.

2.1 Group Recommender System

The emerging demand to provide group recommendations led to the establishment of GRS. GRS [3, 14] methods often adhere to a three-step procedure: (a) group formation – forming group members by identifying users with similar preferences, (b) group modeling – aggregating group members' preferences, and (c) prediction – predicting the unrated items. Within the framework of GRS, groups can be classified in a variety of ways based on various member-related attributes, such as the types of preferences or the cause for the group's establishment [15].

2.2 Cross-Domain Recommender System

Cross-domain methods [5–7] have largely been studied to enhance recommendations in a target domain with insufficient user preferences. The cross-domain approach is one practical solution to addressing RS's data sparsity and cold-start issues [7]. Transferring knowledge from the source domain to enhance the knowledge in the target domain is a popular cross-domain approach to reducing data sparsity issues. By taking advantage of item attributes and user preferences in several domains that are linked to the target domain, it aims to overcome the data shortage.

2.3 Linked Open Data

LOD [3, 8] is a successful manifestation of data links on the Web. It combines disparate data from multiple sources across businesses to create new knowledge and allow sophisticated services and applications. Linked data assets in the so-called "LOD Cloud" [16] were set up to publish graph-shaped data assets in an openly accessible manner using standard Web protocols [16]. Data publishers from various fields have released many datasets based on LOD principles over the past ten years.

3 Methodology

Figure 1 introduces our proposed model in a research framework used in this study to develop a cross-linked domain model that utilizes LOD technology and cross-domain methods simultaneously, as discussed in the prior sections, respectively, to generate enhanced data in the target domain. It consists of mapping linked datasets through this model and its potential for future implementation in collaborative GRS.

As a reference shown in the top left of Fig. 1, we determined Dataset 1 from the movie domain as the target domain, intending to enrich the original sparse movie dataset into an enhanced dataset with lower sparsity through the proposed model. In contrast, Dataset 2 from the music domain will be the source domain in this study. The extracted auxiliary information from the music domain is aimed to transfer to the movie target domain. To leverage LOD technology, we used DBpedia for data extraction through the linked datasets on its platform using 'SPARQL Protocol and RDF Query Language' (SPARQL). DBpedia represents extracted information using the Resource Description Framework (RDF). RDF is used to combine data from several sources and domains

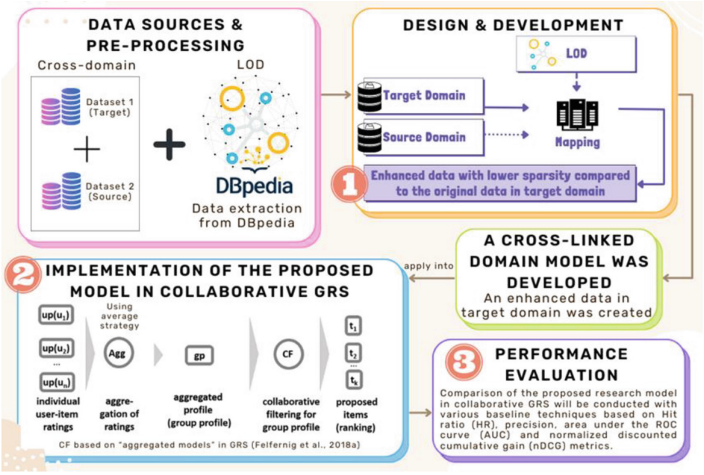

Fig. 1. Research framework.

on DBpedia. Hence, SPARQL can extract the linked data from DBpedia, one of the platforms in the LOD cloud.

We identified the mapping relation between the chosen movie dataset and the linked datasets on DBpedia to pull the auxiliary information from both the movie and music domains into the enhanced dataset in the target domain, as shown in the first phase, in the top right of Fig. 1. This is to generate an enriched dataset with more helpful information to achieve a higher prediction accuracy. Transferring extracted auxiliary data into the movie domain can alleviate the sparsity in a user-movie matrix.

This study developed a cross-linked domain model that leverages LOD technology and cross-domain simultaneously to extract auxiliary information from another domain and evaluate its performance in collaborative individual RS. This ensures that the data sparsity can be alleviated before group formation using individual user profiles.

The subsequent phases of implementation of our proposed model and its performance evaluation, as shown at the bottom of Fig. 1, will be discussed in Sect. 5: Conclusion and Recommendation. This study will end before the second phase, as shown in Fig. 1, by proving that the enhanced data can accurately predict the individual ratings in the sparse user-item matrix. The following experiment in this study will evaluate whether the proposed model can alleviate the data sparsity in individual user profiles before group formation, assuming that addressing data sparsity at the individual level would be more efficient.

3.1 Experiment Setup

As discussed, we classified music as the source domain and movies as a sparse target domain in this experiment. We retrieved the Movie Lens 1 million rating dataset[1] online to evaluate the performance of the proposed method in collaborative individual RS using

[1] Movie Lens 1 million dataset can be retrieved from https://grouplens.org/datasets/movielens/.

the singular value decomposition (SVD) algorithm as a matrix factorization approach to study underlying patterns and interactions in the user-movie matrix. We sampled 10% from the 1 million rows in the original dataset into 100 K rows to avoid easy memory crashes due to the limited computational access. We converted the raw data with movie items, users, and ratings into a user-movie matrix.

The sparsity of this sampled 100 K dataset is 0.9949. The sparsity divides the number of zero elements in the user-movie matrix by the total number of elements in the matrix as the formula (1) shown below. The result will be a decimal between 0 and 1, where 0 indicates no sparsity (all elements are non-zero), and 1 indicates complete sparsity (all elements are zero). The formula for sparsity is:

$$\text{Sparsity} = \text{Number of zero elements}/\text{Total number of elements} \tag{1}$$

We used DBpedia to extract and collect additional LOD features in this study. We utilized the linked datasets from DBpedia to extract auxiliary music and movie features. DBpedia is one of the LOD platforms that links movies and related information in the source and target domains, which are the music and movie domains in this study, respectively.

To create the model, domain item information was extracted from the linked datasets, which link to items and concepts in the source and target domains. This operation needs data mapping to obtain the data from the LOD dataset. Thus, it must recognize bridges as a strategy for collecting LOD information and transferring information across domains. The objective was to alleviate the sparse target domain by examining a prospective collaborator represented by a collection of users or items from a source domain. This study used the 'movie title' feature as the bridge between Movie Lens 1 million and linked datasets.

Since the movie titles were not matched between these two datasets, we used the mapping dataset by Noia et al. [17, 18] to map the 'title' feature in the MovieLens 1M dataset to the DBpedia for data extraction. As the data in the movie target domain is limited, we utilized LOD technology to collect and extract additional movie-related features of 'director' and 'starring' that are considered in the same movie domain. However, the available helpful feature is also limited to a specific domain.

Given the ideas, we leveraged LOD technology to collect and extract the 'music composer' feature from another domain. Hence, we utilized the cross-domain feature in this study of using 'music composers' of the movies in the music domain. We then identified the 'music composer' of the movies as auxiliary information in the source domain. We assumed that if someone likes music composed by their beloved composer, they might watch the movie and have a higher chance of loving it. For example, John William is one of the famous music composers who wrote the theme series for most Star Wars movies. It is a simple instance that the user has a higher chance of loving all Star Wars movies. A similar assumption applies to 'starring' and 'director' features.

Algorithm 1 in Fig. 2 forms an enhanced trainset to train our proposed model with additional LOD and cross-domain information. For this study, directors who have directed at least three movies and actors who have starred in at least three movies for all rows in the Movie Lens dataset were extracted, respectively.

As shown in Fig. 2, a sample of 100K rows from the enhanced Movie Lens 1 million dataset with three additional feature information was created as input for the below

Algorithm 1. A user-movie matrix was formed using a pivot table. The "Not a Number" (NaN) were missing values in Python and filled with 0, assuming NaN means the user has not rated the movie. The 'Music Composer' and 'Director' feature information were encoded using one-hot encoding, while 'Starring' was converted into a binary representation using the respective method. This is to combine the encoded LOD and cross-domain feature information with the user-movie matrix to build an enhanced trainset as the output of this algorithm.

Algorithm 1

```
Input: Sampled 100K dataset from enhanced Movie Lens 1
million data with three features ('director'& 'starring'
>= in 3 movies and respective 'music composer')
Output: Enhanced trainset of the extended user-movie
matrix with LOD and cross-domain information
START
   1. Create a pivot table (matrix) from the input
      dataset
   2. Handle and fill NaN values with 0 in the 'Music
      Composer,' 'Director,' and 'Starring' feature
      columns
   3. One-hot encode 'Music Composer' and 'Director';
      encoder_composer = OneHotEncoder(sparse=False)
      encoder_director = OneHotEncoder(sparse=False)
   4. Convert 'Starring' to binary representation;
      mlb_starring = MultiLabelBinarizer()
   5. Combine the encoded features with the user-movie
      matrix
   6. Build a trainset from the combined dataset;
      trainset_with_features =
      data_with_features.build_full_trainset()

END
```

Fig. 2. Algorithm 1 shows the pseudo-code we used to form an enhanced trainset to train our proposed cross-linked domain model at the initial stage.

4 Result and Discussion

The enriched data is then shown via a graph visualization using Neo4j. Program Code 1 showed how the cypher queries were used to split actor names and add 'starring' nodes to link to the movie nodes.

```
FOREACH (starringName IN split(row.Starring, ',') |
   MERGE (s:Starring {name: trim(starringName)})
   MERGE (s)-[:STARRED_IN]->(m)
)
```

Program Code 1. Cypher queries in linking the movie nodes.

The graph visualization in Fig. 3 showed the data linkage between the dataset of Movielens 1M and DBpedia. We can see the nodes and the relationship between three additional linked attributes with cross-domain from the movie and music in this study ('starring,' 'director,' and 'music composer'). This figure illustrated that the cross-domain feature from the music domain can link between different subsets of the movie features. It demonstrated another underlying pattern of user's preference for the movies.

Fig. 3. Linkage of nodes in graph visualization.

This study found similar users by identifying the underlying pattern through latent factors in matrix factorization. This study used collaborative filtering of the SVD algorithm as the matrix factorization method. The user-movie matrix will then be a lower dimension, which can identify latent factors in the model training.

We adopted 'root mean square error (RMSE),' 'mean square error (MSE),' and 'mean absolute error (MAE)' as the fundamental evaluation metrics to ensure diversity in the evaluation process of the trained models to test the performance of our proposed method. The lower the errors, the higher the accuracy of predictions compared to the actual ratings.

We first evaluated the original user-movie matrix without additional features on the linked data using the SVD algorithm, as shown below in Row 1 in Table 2. The prediction errors were among the highest compared to the other methods that used an enriched user-movie matrix, which indicates that this accuracy is poorer in this study.

Moreover, we evaluated the enriched user-movie matrix with each auxiliary LOD and cross-domain feature using a similar SVD algorithm as matrix factorization. The linked

data lowered the error and increased the accuracy of rating predictions compared to the previously mentioned results. The result in Table 2 indicated that using the enriched matrix with only one auxiliary LOD or cross-domain feature was not significant enough to lower the prediction errors, as shown in Row 2–4. Even considering two LOD features of 'director' and 'starring' simultaneously from the same movie domain during the model training, the results were still not in favor, as the errors did not decrease significantly, as shown in Row 5.

Table 2. Experiment results of model testing using different original and enriched datasets with LOD technology and cross-domain based on RMSE, MSE, and MAE.

Method	RMSE	MSE	MAE
1. SVD by using original user-movie matrix from MovieLens 1M	0.9455	0.8939	0.7520
2. SVD-LOD by using an enhanced matrix with a 'director' feature	0.7016	0.4923	0.5575
3. SVD-LOD by using an enhanced matrix with a 'starring' feature	0.7019	0.4926	0.5587
4. SVD-CDLOD by using an enhanced matrix with a 'music composer' feature from the other music domain	0.7021	0.4929	0.5592
5. SVD-LOD by using an enhanced matrix with both 'director' and 'starring' features from the same movie domain	0.7009	0.4913	0.5581
6. **SVD-CDLOD by using an enhanced matrix that leveraged cross-domain information with three 'director,' 'starring,' and 'music composer' features**	**0.5303**	**0.2813**	**0.4223**

As shown in Table 2, this study's proposed method of simultaneously utilizing LOD technology with cross-domain features achieved the lowest errors. It indicated the highest accuracy using the experiment's three 'director,' 'starring,' and 'music composer' features, as shown in Row 6. This suggested that our proposed method of using enriched data to make rating predictions was superior to using sparse data given as the original unenriched dataset. Hence, the enhanced data can then be used to form a user profile for further group formation to provide recommendations to groups.

We experimented with some combination of the auxiliary LOD and cross-domain features related to the patterns of user-movie relationships during the model training using SVD. This aimed to lower the errors between predicted and actual ratings in the movie target domain.

The results strongly suggested that using auxiliary LOD features can lower the prediction errors compared to the actual ratings. Hence, the performance of the recommendation was better and more accurate. The errors were even lower after using the cross-domain and LOD features simultaneously rather than those from a single domain. The result suggested a more accurate prediction of the missing ratings in the original dataset to alleviate sparsity in individual user profiles before group formation. Group formation and implementation into GRS will be carried out in the future.

5 Conclusion and Recommendation

Grouping and clustering are significantly impacted by the sparsity of the data. While many efforts have been made to minimize data sparsity issues in RS for individuals, the impact on GRS remains a significant concern. GRS struggles to deliver reliable recommendations when user profile data is sparse. Our proposed method eliminated data sparsity in user profiles to provide more effective group recommendations. As shown in Fig. 1, the model related to a cross-linked domain will subsequently be implemented in the collaborative GRS with the selected aggregation strategy. Future research will build on this study's findings by including social metadata in cross-domain recommendations. Additionally, algorithms for ranking and filtering objects in the target domain by computing semantic similarities will be created. Comparison of the proposed research model in collaborative GRS will be subsequently conducted with various baseline techniques based on 'hit ratio (HR),' 'precision,' 'area under the ROC curve (AUC),' and 'normalized discounted cumulative gain (nDCG)' metrics.

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